

DATA ANALYTICS PORTFOLIO PROJECT

Customer Churn Analysis

An End-to-End Data Analytics Project Using Python, MySQL & Power BI

Analyzing customer behavior, churn patterns, tenure, revenue and customer-support activity.

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Objective, Dataset & Workflow

Project Objective

Analyze customer churn and identify observable patterns across:

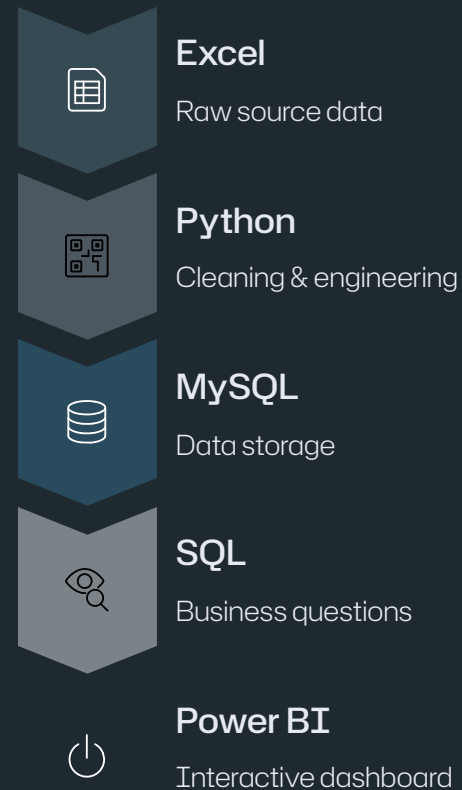
- Customer demographics & geography
- Subscription & plan type
- Tenure & revenue
- Complaints & CSAT

Dataset

Table	Rows
db_customer	521
db_subscription	521
db_support	222

Final analytical dataset: **521 customers**

End-to-End Workflow



PYTHON STAGE

Data Cleaning & Feature Engineering

Data Cleaning

- Data inspection & missing-value analysis
- Duplicate checking & date conversion
- Removal of unnecessary fields
- Customer-level merging & support-data aggregation

Feature Engineering

- Age, Age group, Churn flag
 - Tenure days, Tenure years, Tenure group
 - Annualized charges, Cancellation year/month
 - Churn risk group, Escalation flag
- Complaint count, Escalation count, Avg. CSAT

Churn Overview & SQL Analysis

521

Total Customers

Full dataset

137

Churned

Lost customers

384

Active

Retained customers

26.3%

Churn Rate

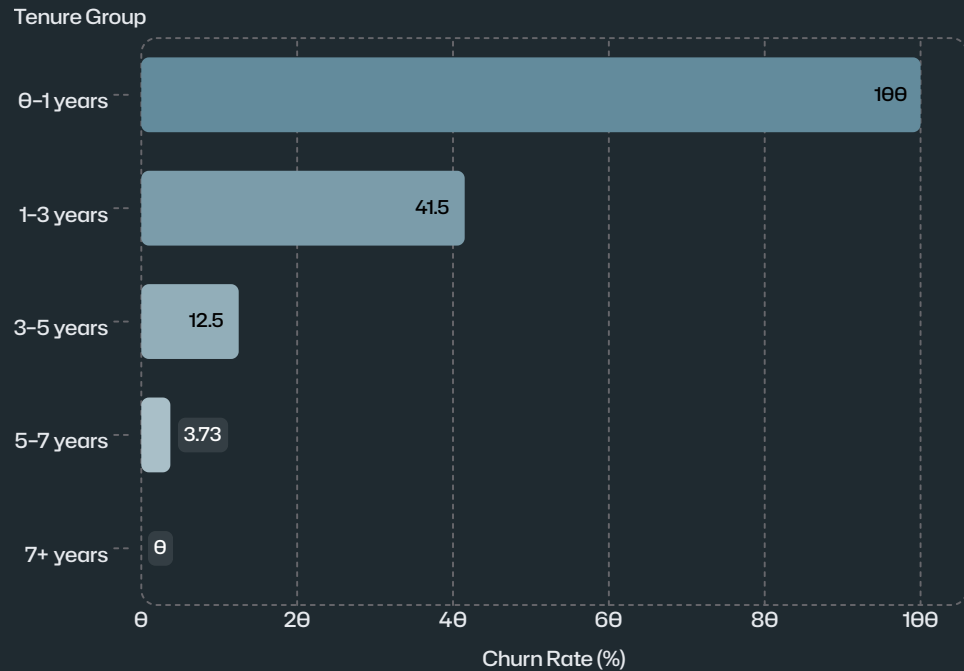
Overall observed

MySQL was used to store the cleaned dataset and SQL was used to answer **12 business questions** across the analytical dataset.

Database	Table
customer_churn_analysis	customer_churn_data

Tenure & Churn Analysis

Churn Rate by Tenure Group



Average Tenure by Status

4.64

Active (yrs)

Average tenure

1.68

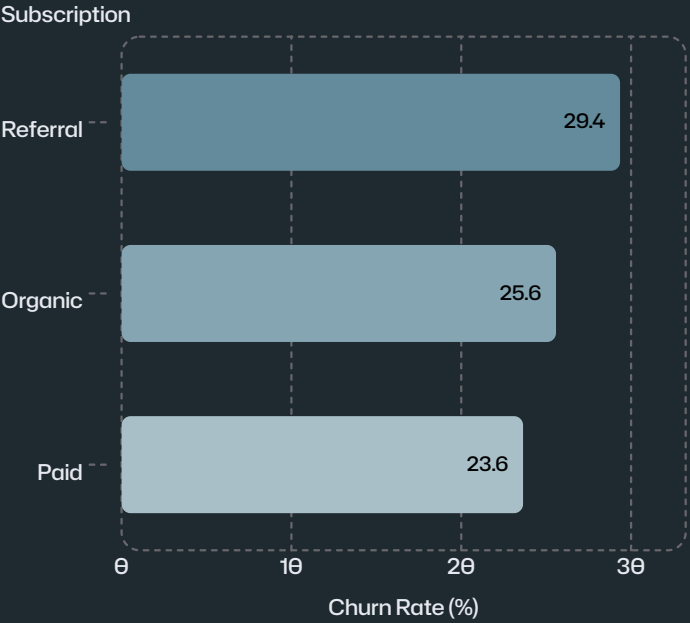
Churned (yrs)

Average tenure

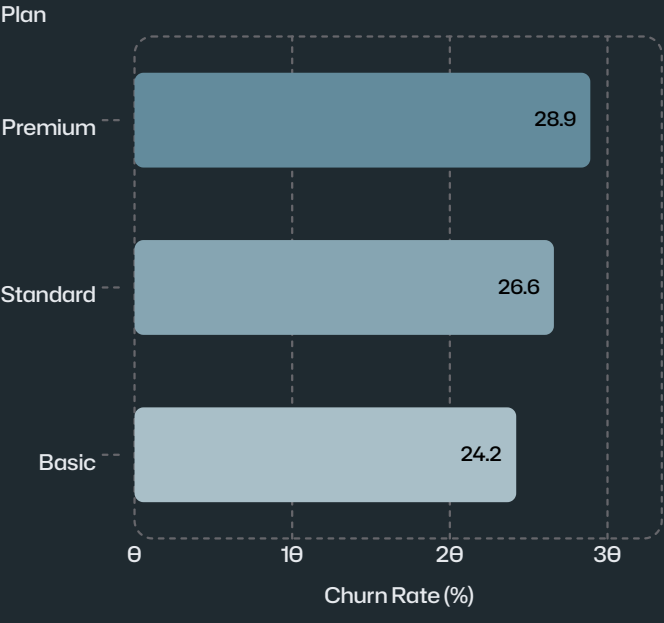
❏ **Note:** Extreme values in small tenure groups should be interpreted carefully. Shorter-tenure customers show substantially higher observed churn in this dataset.

Subscription, Plan & Age Group Analysis

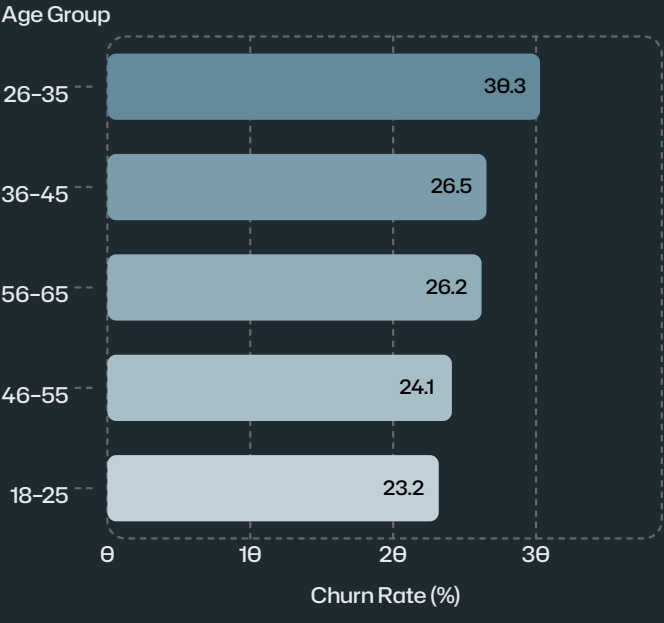
By Subscription Type



By Plan Type

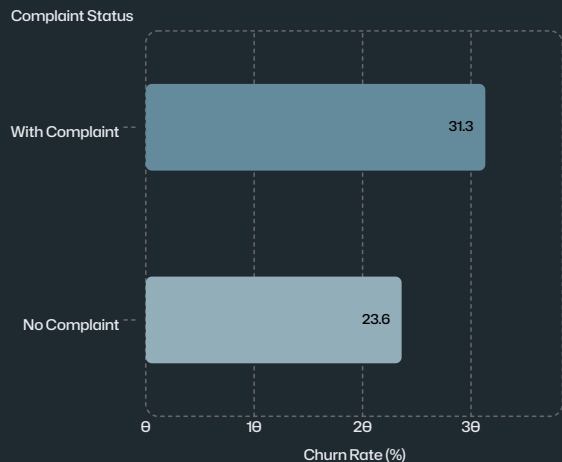


By Age Group



Customer Support, Revenue & Geography

Complaint Analysis



38.46

Average CSAT

Churned customers

35.98

Average CSAT

Active customers

Complaint analysis shows association, not causation. CSAT covers customers with available support records.

Revenue (Annualized: Monthly $\times$ 12)

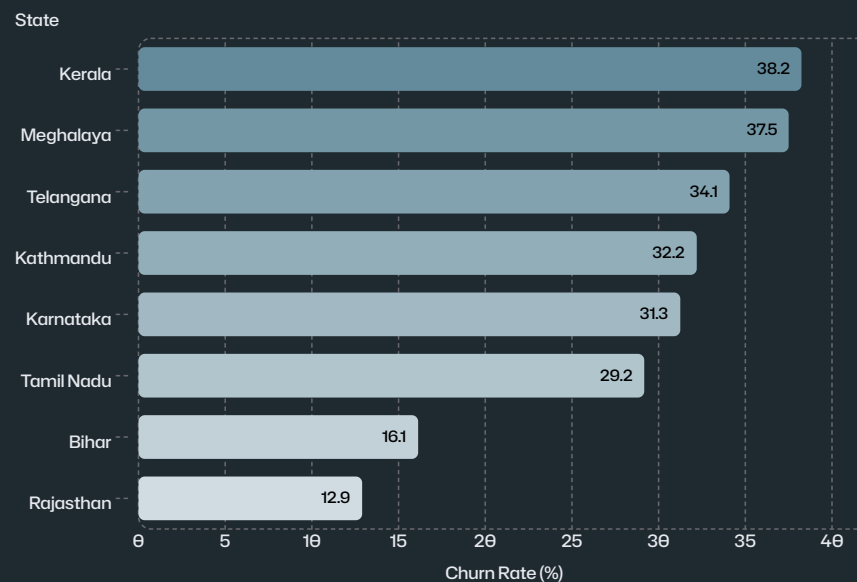
Active Customers

68,831.04

Churned Customers

24,872.64

State-Level Observed Churn (Selected)



State-level differences should be interpreted alongside the number of customers in each state.

Power BI Dashboard – 5 Pages

1. Churn Overview

KPIs, monthly churn trend,
contract type and plan type

2. Customer Demographics

Gender, country, age group
and geography

3. Subscription & Tenure

Tenure, subscription type,
plan and contract analysis

4. Revenue & Churn Risk

Revenue, churned revenue,
revenue at risk and churn-risk
groups

5. Customer Support

Complaints, escalations, CSAT and support-related churn

"The dashboard allows interactive filtering and exploration of customer churn across multiple business dimensions."

KEY FINDINGS

Business Insights & Key Findings

Overall Churn – 26.39%

Observed churn across 521 customers.

Tenure – 1.68 vs 4.64 yrs

Avg tenure of churned vs active customers.

Early Tenure – 41.59%

Observed churn in the 1-3 year tenure group.

Complaints – 31.32% vs 23.69%

Observed churn with vs without recorded complaints.

Revenue at Risk – 24,872.64

Annualized charges associated with churned customers.

Areas for Further Investigation

- Early-Tenure Onboarding
- Complaint Resolution
- Customer Engagement
- Revenue Exposure
- Support Experience

Conclusion

This project demonstrates a complete customer churn analytics workflow from raw Excel data to SQL analysis and an interactive Power BI dashboard.

Python

Cleaning + Feature Engineering

MySQL / SQL

Data Storage + Business Analysis

Power BI

Visualization + Interactive Reporting

Limitations

- Dataset contains 521 customers
- Some segments may have few customers
- Analysis is descriptive, not predictive
- Complaints show association, not causation
- Revenue based on monthly charges $\times$ 12
- CSAT covers available support records only
- Historical patterns $\neq$ future predictions

Project Architecture

01

Excel

Raw source data

02

Python

Cleaning & engineering

03

MySQL $\rightarrow$ SQL

Database & business questions

04

Power BI

Interactive dashboard

05

Business Insights

Findings & recommendations

Thank You

Customer Churn Analysis | Python • MySQL • SQL • Power BI

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